

Fourier Transforms with Potential Fields

Instructor Key and Interpretation Guide

Geophysics IIIA
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Purpose of This Document

This document provides an instructor-only interpretation guide and marking key for the Jupyter notebook practical on Fourier transforms applied to gravity and magnetic data from South Australia. It is intended to clarify expected observations, physical interpretations, and common student misconceptions. Students are not expected to provide all of the detail included here; rather, this guide defines what constitutes a sound geophysical understanding for each question.

Notebook Structure

The practical notebook (`week_7_fourier.ipynb`) is organised into six sections:

1. Load and display gravity, magnetics, and topography/sediment grids
2. Interactive profile selection (click on a map, or use fallback coordinates)
3. One-dimensional profile continuation with an interactive slider
4. Two-dimensional gravity continuation and isostatic comparison
5. Reduction to the pole (interactive inclination and declination sliders)
6. Sediment-thickness correction for RTP (*optional/exploratory*)

The problem set (`week7_problem_set.tex`) mirrors this structure with 20 questions grouped into five assessed sections plus one optional section.

Learning Outcomes

By the end of this practical, students should be able to:

- Interpret spatial and spectral representations of potential-field data
- Explain upward and downward continuation as frequency-domain operations
- Distinguish regional and residual fields and critically evaluate the choices involved
- Compare spectral and isostatic approaches to defining a regional gravity field
- Apply reduction to the pole and understand its assumptions and limitations

- Discuss how sediment thickness affects the interpretation of magnetic data
- Appreciate the non-uniqueness and ill-posed nature of potential-field interpretation

1 Section 1 — One-Dimensional Profile Continuation (Q1–5)

Students extract co-located gravity and magnetic profiles along a user-selected great-circle path, then apply the 1-D continuation filter $H(k, z) = e^{-2\pi|k|z}$ (wavenumber in cycles km^{-1} , height z in km) using an interactive slider from -20 to $+100$ km.

Q1 — Upward continuation as a low-pass filter

Expected answer. Narrow, localised anomalies disappear first because they correspond to high spatial wavenumbers; the exponential decay factor $e^{-2\pi|k|z}$ is smallest for large $|k|$. Broad, regional features persist longest because they are associated with small $|k|$. This implies that short-wavelength anomalies originate from shallow sources, while long-wavelength anomalies reflect deeper or more laterally extensive structure.

Common misconceptions.

- Claiming that broad anomalies disappear first.
- Confusing amplitude attenuation with frequency-independent smoothing.

Q2 — Residual anomalies

Expected answer. The residual (observed – regional) emphasises sources not accounted for by the long-wavelength regional, typically shallower lithological contrasts, intrusive bodies, or structural offsets. Gravity and magnetic residuals are often spatially correlated where lithology governs both density and susceptibility (e.g. mafic intrusions), but may disagree where remanent magnetisation decouples the two fields, or where a density contrast exists without a corresponding susceptibility contrast.

Q3 — Choosing a continuation height

Expected answer.

- Defensible criteria include: visual inspection (the height at which known deep regional features become smooth while local anomalies remain visible in the residual); comparison with independent depth estimates (e.g. Moho depth from seismic); spectral depth analysis (slope of the log-power spectrum); or geological judgement about the expected crustal structure.
- Sensitivity is typically high. Students should observe that the character of the residual changes substantially over a range of only a few kilometres, which is a key point about the subjective nature of the regional–residual separation.

Q4 — Instability of downward continuation

Expected answer.

- Visible instability (ringing, oscillations, extreme amplitudes) typically sets in within a few kilometres of downward continuation. The exact depth depends on the noise level of the dataset.

- b. Magnetic data tend to become unstable at shallower depths than gravity because magnetic datasets typically have higher spatial frequency content and more noise.
- c. The filter $e^{+2\pi|k||z|}$ amplifies all wavenumbers; high-wavenumber noise grows fastest. The effective maximum stable depth is set by the noise floor of the measurement, which differs between the two datasets.

Q5 — Physical meaning of the continued field

Expected answer. Yes, the upward-continued profile should look physically reasonable: it represents the field that would be measured on a higher observation surface. It should be smoother and lower in amplitude. Features that seem unreasonable (e.g. negative-amplitude oscillations flanking a positive body) may reflect edge effects from the FFT (wrap-around), or the presence of sources that violate the harmonic-function assumption (e.g. sources within the continuation region).

2 Section 2 — 2-D Gravity Continuation and Isostasy (Q6–9)

The 2-D continuation filter $H = e^{-2\pi\sqrt{k_x^2+k_y^2}z}$ is applied to the full gravity grid with a fixed reference height of 25 km for the comparison maps. An Airy isostatic correction is computed as $g_{\text{iso}} = 2\pi G \rho_c h$, where h is positive topography (sub-sea-level values are zeroed).

Q6 — Spectral regional vs isostatic correction

Expected answer. The two fields share long-wavelength trends but differ in detail.

- a. Broad agreement is expected in elevated terrain where Airy compensation is a reasonable approximation.
- b. The largest differences occur in areas of thick sedimentary basins (which have near-zero topography but significant subsurface density variation), along the coast (ocean loading is not captured), and in regions of tectonic flexure where isostasy is not locally achieved.

Q7 — Residual comparison and compensation depth

Expected answer. Students should find rough visual agreement at continuation heights in the range of 30–50 km, consistent with compensation depths for cratonic crust. This does not prove compensation depth equals that height; it reflects the scale of the features that Airy compensation removes.

Q8 — Anomalies not explained by isostasy

Expected answer. Anomalies in the spectral residual but absent from the isostatic residual reflect features accounted for by upward continuation but not by Airy topographic compensation, e.g. deep crustal density contrasts, underplated mafic material, or lateral mantle heterogeneity. The converse (present in isostatic but not spectral residual) may indicate shallow features whose gravity is removed by continuation but not by the topographic model.

Q9 — Limitations of the Airy model

Expected answer. Key assumptions and their failure modes:

- *Local compensation*: flexural rigidity of the lithosphere means loads are regionally compensated; the Airy model therefore over-corrects beneath narrow ranges and under-corrects adjacent to them.
- *Uniform crustal density ρ_c* : lateral variations in crustal composition are ignored.
- *Topography as the only load*: sediment thickness, ice loading, and internal density anomalies are not accounted for.
- *Flat Moho geometry*: the model assumes a planar reference Moho, ignoring pre-existing crustal structure.

3 Section 3 — Reduction to the Pole (Q10–13)

The stabilised RTP filter $H_{\text{RTP}} = \tilde{\Theta}^2 / (|\Theta|^2 + \varepsilon)^2$ is applied to the total magnetic intensity (TMI) grid. Physical spacing in km is used for the wavenumber axes. Students explore inclination (-90° to -10°) and declination (-30° to $+30^\circ$) interactively.

Q10 — Effect on anomaly positions

Expected answer. In the Southern Hemisphere, positive TMI anomalies are displaced southward from their sources due to the negative inclination of the ambient field. RTP shifts anomaly maxima northward, centring them above the causative bodies. The shift magnitude depends on inclination; at $I \approx -60^\circ$ (typical for South Australia) the displacement is moderate, of order 5–20 km for typical anomaly widths. Shape changes from asymmetric (TMI) to more symmetric (RTP) should also be noted.

Q11 — Stability at low inclinations

Expected answer.

- Artefacts typically appear at inclinations shallower than about $\pm 15^\circ$ to $\pm 20^\circ$, though the water-level parameter $\varepsilon = 0.05$ provides some stability. Students should see ringing and extreme amplitudes in the RTP map as the slider approaches 0° .
- Near the equator ($I \approx 0$), the direction-cosine factor Θ approaches zero for wavenumbers in the north–south direction, making the filter singular. The inversion is numerically unstable even with regularisation, so RTP cannot be reliably applied at low magnetic latitudes.

Q12 — Difference map (RTP – TMI)

Expected answer. The difference map highlights the phase correction — positive differences appear to the north of anomaly centres (where RTP has moved the peak) and negative differences to the south. This dipolar difference pattern encodes the along-field asymmetry of the original TMI anomaly. Larger differences indicate either a stronger phase shift (lower inclination, shallower source) or stronger magnetisation contrast.

Q13 — Remanent magnetisation

Expected answer. The RTP filter applies a single phase correction based on the ambient-field direction. Where a body carries remanent magnetisation in a significantly different direction (e.g. a reversely magnetised unit), RTP will *over-correct* or *under-correct* the anomaly, potentially distorting its shape and position

rather than centring it. Students who have covered remanence (e.g. in Week 6) may identify candidate anomalies as those that do not become symmetric after RTP, or that show unexpected polarity.

4 Section 4 — Sediment-Thickness Correction (Optional, Q14–17)

This section is explicitly marked optional in the problem set. It is intended as an exploratory exercise. The combined filter applies RTP plus a downward continuation of z_{sed} km damped by a Gaussian taper of length scale λ km.

Q14 — Where the correction matters most

Expected answer. The largest differences between standard and sediment-corrected RTP should appear in regions of thickest sediment (the Officer Basin, Murray Basin, and Eucla Basin). Students should verify this against the sediment-thickness map shown in Section 1 of the notebook.

Q15 — Role of the damping parameter λ

Expected answer. Increasing λ (the Gaussian taper length scale) progressively damps the short-wavelength amplification of the sediment correction. At large λ , the correction effectively removes the sediment-depth inversion entirely for high wavenumbers, leaving only the long-wavelength component. Whether this is geologically sensible depends on the data noise level; students should observe that large λ produces a smooth but geologically uninformative correction, while small λ produces artefacts.

Q16 — Validity of the uniform approximation

Expected answer.

- The approximation is most valid where sediment thickness is close to the mean value, i.e. in moderately covered areas away from basin margins. It is least valid in the centres of thick basins (over-correction) and in exposed basement terranes (unnecessary correction).
- A rigorous spatially variable correction would require either a depth-to-basement model at grid resolution, a tiled spectral approach (windowed FFT with spatial overlap), or iterative modelling.

Q17 — Analogous gravity correction

Expected answer. A downward-continuation gravity correction for sediment would attempt to recover the gravity field at the basement surface rather than the observation surface. It would differ from the Airy correction in that: (1) the Airy correction removes the long-wavelength gravity effect of the isostatic root driven by topography, whereas a sediment downward continuation attempts to resolve the gravity field at depth; and (2) a downward continuation is inherently unstable, while the Airy correction is a stable forward model. Both approaches are geologically approximate.

5 Section 5 — Synthesis (Q18–20)

Q18 — Wavelength and depth

High-quality answer. The approximate relationship is $\lambda \approx 2\pi d$, where λ is the dominant wavelength and d is the source depth (derived from the radial decay of a point source). Key assumptions are: isolated,

compact sources; source-free continuation region; and stationary noise statistics. A misleading depth estimate arises when a large, laterally extensive source (e.g. a thick sedimentary basin) dominates the long-wavelength signal — the “depth” inferred from the spectral slope would reflect the lateral extent of the source, not its burial depth.

Q19 — Spectral vs isostatic regional field

High-quality answer. *Spectral continuation* is mathematically objective and requires no geological model, but the choice of height is arbitrary and geologically unmotivated; it works well for separating sources at different depths. *Isostatic correction* is physically motivated and removes the predicted gravity of crustal roots, but assumes a specific compensation mechanism (Airy), requires density parameters, and fails in regions of flexural or dynamic topography. The spectral approach is preferred for exploratory analysis; the isostatic approach is preferred where the geological setting supports local Airy compensation and the goal is to isolate intra-crustal anomalies.

Q20 — Non-uniqueness

High-quality answer. A concrete example: at a suitable continuation height, the spectral residual gravity map closely resembles the isostatic residual gravity map, yet the two are derived from entirely different assumptions (mathematical filter vs. physical compensation model). Both “explain” the observations, but they imply different geological models. Discriminating between them would require independent constraints such as seismic refraction or reflection profiles (to determine actual Moho depth and crustal thickness), well data (to constrain density and susceptibility), or receiver function analysis.

Marking Guidance

- **Full marks** require correct physical reasoning, not just description of what is seen on screen.
- **Partial marks** for correct observation without physical explanation, or correct explanation applied to an incorrect observation.
- **No marks** for purely descriptive statements that restate the question (e.g. “the continued field is smoother because continuation makes it smoother”).
- The synthesis questions (Q18–20) are worth more and should demonstrate integration across sections. Accept well-reasoned answers that differ from the expected answer provided the geophysical logic is sound.
- The optional sediment correction section (Q14–17) is marked generously given its exploratory nature; acknowledge any thoughtful engagement with the instability and approximation issues.

End of Instructor Key